

# 2026 ESC Guidelines for the management of cardiovascular disease and chronic kidney disease

**Official Slide set**

# Slide set history

The material was adapted from the '*2026 ESC Guidelines for the management of cardiovascular disease and chronic kidney disease, in collaboration with the European Renal Association (ERA)*' (*European Heart Journal*; 2026 - doi: 10.1093/eurheartj/ehag098) as published on 28/08/26.

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# ESC Classes of recommendations

| Class of recommendation | Definition                                                                                                                     | Wording to use                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>I</b>                | Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.                        | <i>“is recommended”<br/>or “is indicated”</i> |
| <b>II</b>               | Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.         |                                               |
| <b>IIa</b>              | Weight of evidence/opinion is in favour of usefulness/efficacy.                                                                | <i>“should be considered”</i>                 |
| <b>IIb</b>              | Usefulness/efficacy is less well established by evidence/opinion.                                                              | <i>“may be considered”</i>                    |
| <b>III</b>              | Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful. | <i>“is not recommended”</i>                   |

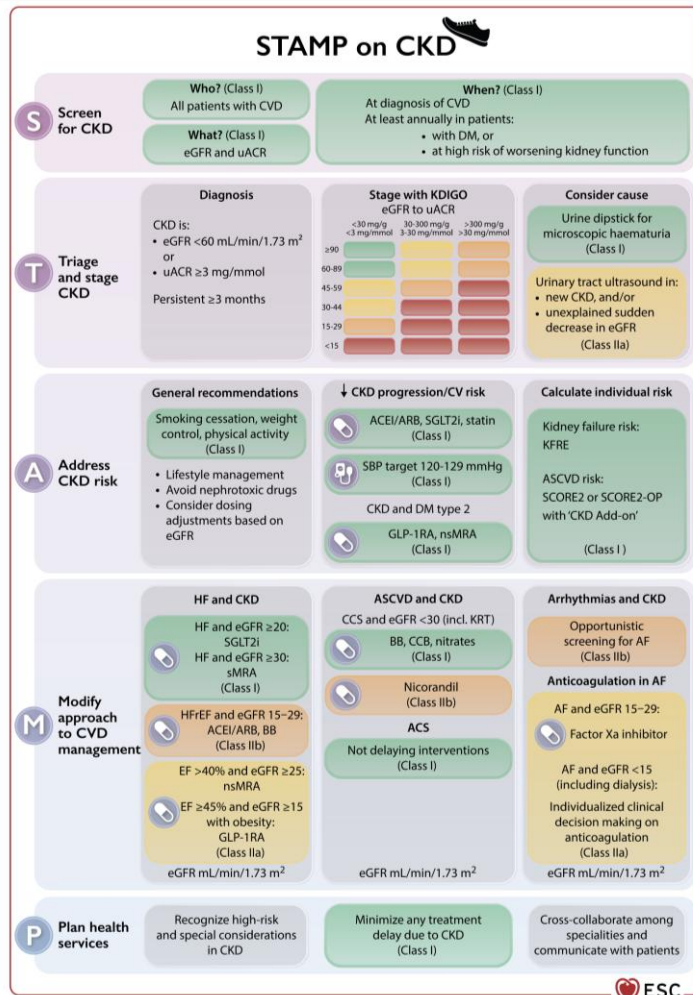
# ESC Levels of evidence: therapy and prevention

| Level of evidence | Definition                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b>          | Conclusive evidence, usually from at least two adequately powered randomized controlled trials <sup>a</sup> free of major bias, with substantial evidence against the play of chance when combined in a meta-analysis (e.g., $P < 0.005$ for superiority). <sup>b</sup>                                                                                                                                     |
| <b>B1</b>         | Suggestive evidence from at least one adequately powered randomized controlled trial free of major bias, or a meta-analysis of such randomized controlled trials, with some evidence against the play of chance (e.g., $P < 0.05$ for superiority). <sup>c</sup>                                                                                                                                            |
| <b>B2</b>         | Limited evidence from at least two adequately powered non-randomized studies with careful control of major sources of bias and substantial evidence against the play of chance (e.g., $P < 0.005$ for superiority), <sup>b</sup> or from a meta-analysis of small, underpowered randomized controlled trials with some evidence against the play of chance (e.g., $P < 0.05$ for superiority). <sup>c</sup> |
| <b>C</b>          | Preliminary evidence from non-randomized studies without careful control of major sources of bias, a single small, underpowered randomized controlled trial, or expert consensus.                                                                                                                                                                                                                           |

# ESC Levels of evidence: diagnostic tests and predictions models

| Level of evidence | Diagnostic test                                                                                                                                   | Prediction model<br>(including diagnostic and prognostic models)                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b>          | Conclusive evidence of adequate diagnostic ability from at least two high-quality studies                                                         | Conclusive evidence of adequate predictive ability from at least one high-quality derivation study, and two or more external validation studies of at least moderate quality.                                                                        |
| <b>B</b>          | Suggestive evidence of adequate diagnostic ability from a single high-quality or at least two moderate-quality studies.                           | Suggestive evidence of adequate predictive ability from one or more derivation and one or more external validation studies of at least moderate quality.                                                                                             |
| <b>C</b>          | Preliminary evidence of adequate diagnostic ability from a single moderate-quality study, any number of low-quality studies, or expert consensus. | Preliminary evidence of adequate predictive ability from a derivation study of at least moderate quality, but with external validation absent or in a low-quality study, or a derivation study of low quality (irrespective of external validation). |

# Figure 1 Central Illustration



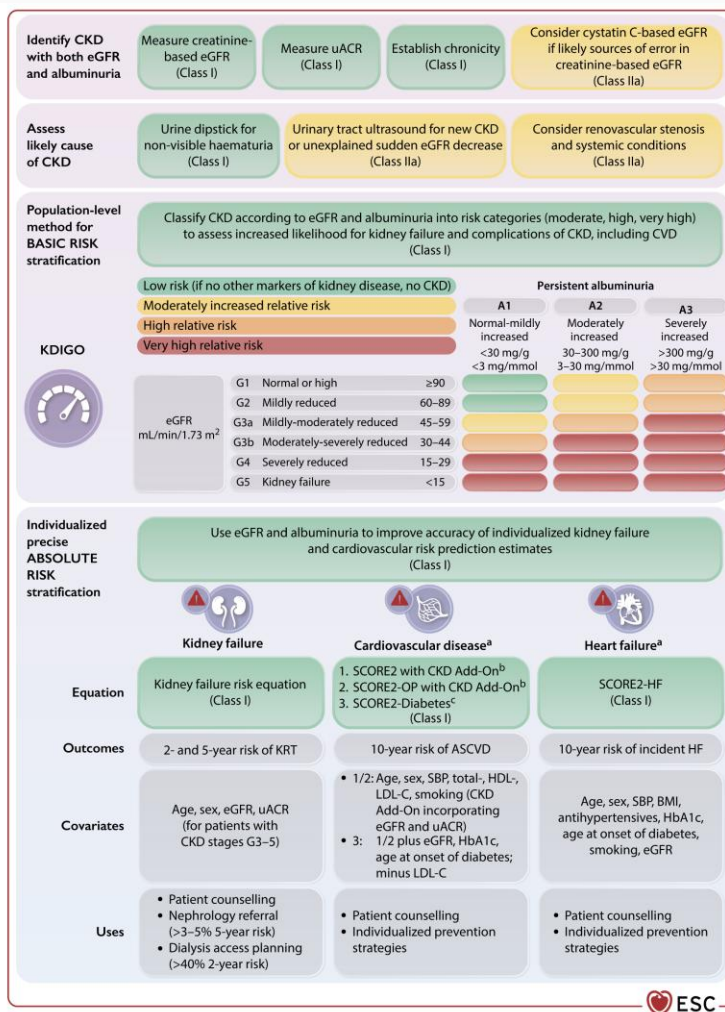
# Figure 2 Recommended medical interventions by diabetes status and different cardiovascular diseases

|             | ACEI/ARB                                   | SGLT2i                                                                            | GLP1RA                                                | sMRA or nsMRA                                               | Beta-blockers                                   | Statin-based regimen                                               | Anti-thrombotic therapy                                                                         |
|-------------|--------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| No DM       | CKD progression, CV events (Class I)       | + albuminuria<br>Kidney failure, CV events (Class I)<br>- albuminuria (Class IIa) |                                                       |                                                             |                                                 | MACE<br>CKD without KRT (Class I)<br>Kidney transplant (Class IIa) | Without ASCVD<br>Risk of bleeding may outweigh CV benefit (Class III)                           |
| DM          | CKD progression, CV events (Class I)       | Kidney failure, CV events (Class I)                                               | + albuminuria<br>CKD progression, CV events (Class I) | + albuminuria<br>Kidney failure, CV events (nsMRA: Class I) |                                                 | MACE<br>CKD without KRT (Class I)<br>Kidney transplant (Class IIa) | Without ASCVD<br>Risk of bleeding may outweigh CV benefit (Class III)                           |
| HF rEF      | Including ARNI<br>HF hosp, death (Class I) | HF hosp, CV death (Class I)                                                       |                                                       | sMRA<br>HF hosp, death (Class I)                            | HF hosp, death (Class I)                        |                                                                    |                                                                                                 |
| HF pEF      |                                            | HF hosp, CV death (Class I)                                                       | EF $\geq 45\%$ and obese<br>Improve QoL (Class IIa)   | EF $\geq 40\%$ (MRA)<br>HF events, CV death (Class IIa)     |                                                 |                                                                    |                                                                                                 |
| ASCVD       |                                            |                                                                                   |                                                       |                                                             | Angina (Class I)                                | MACE<br>CKD not on dialysis (Class I)<br>Dialysis (Class IIb)      | MACE (Class I)                                                                                  |
| Arrhythmias |                                            |                                                                                   |                                                       |                                                             | Rate control in AF (no specific recommendation) |                                                                    | Factor Xa inhibitor in AF<br>eGFR $\geq 30$<br>Stroke, VTE (Class I)<br>eGFR: 15–30 (Class IIb) |

# Recommendations for the diagnosis of chronic kidney disease

| Recommendations                                                                                                                                                                                                                 | Class | Level |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Classifying CKD according to GFR and albuminuria into moderate, high, or very high-risk categories is recommended to assess increased likelihood for kidney failure and complications of CKD, including cardiovascular disease. | I     | A     |
| Measurement of a creatinine-based eGFR using a validated estimating equation is recommended to assess kidney function and to stage CKD.                                                                                         | I     | A     |
| Measuring uACR in a spot morning urine sample is recommended to assess albuminuria and classify CKD stage.                                                                                                                      | I     | A     |
| Two measurements of eGFR and uACR over a period of at least 3 months are recommended to establish chronicity needed to confirm CKD.                                                                                             | I     | C     |
| Measuring cystatin C-based eGFR should be considered in patients with likely sources of error in creatinine-based eGFR when an accurate assessment of kidney function is needed for clinical decision-making.                   | Ila   | A     |

# Figure 3 Chronic kidney disease classification and risk prediction approaches



# Recommendations for the causes of chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                                                 | Class | Level |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Performing a urine dipstick test for microscopic haematuria is recommended in patients with newly identified CKD with the aim of detecting glomerular disease requiring nephrological or urological disease evaluation.                                                         | I     | C     |
| A urinary tract ultrasound should be considered in patients with newly diagnosed CKD and also in patients with an unexplained sudden decrease in eGFR to assess kidney size and symmetry, and to evaluate possible urinary tract obstruction, or other causes of decreased GFR. | Ila   | C     |

## Recommendations for the causes of chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Class      | Level    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| <p>Evaluation for renovascular stenosis with suitable imaging modality should be considered in patients with:</p> <ul style="list-style-type: none"><li>• Rapidly progressive CKD without clear cause</li><li>• <math>\geq 30\%</math> change in eGFR upon starting or stopping a single renal haemodynamically active medication</li><li>• Recurrent flash pulmonary oedema, or</li><li>• Treatment-resistant hypertension on <math>\geq 4</math> appropriately dosed blood pressure-lowering agents</li></ul> <p>to identify patients with CKD with clinically relevant renal artery disease who may benefit from PTR.</p> | <b>IIa</b> | <b>C</b> |
| <p>Assessment for systemic conditions (e.g. amyloidosis or Fabry disease) should be considered in patients with unexplained heart failure and CKD to inform specific management.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>IIa</b> | <b>C</b> |

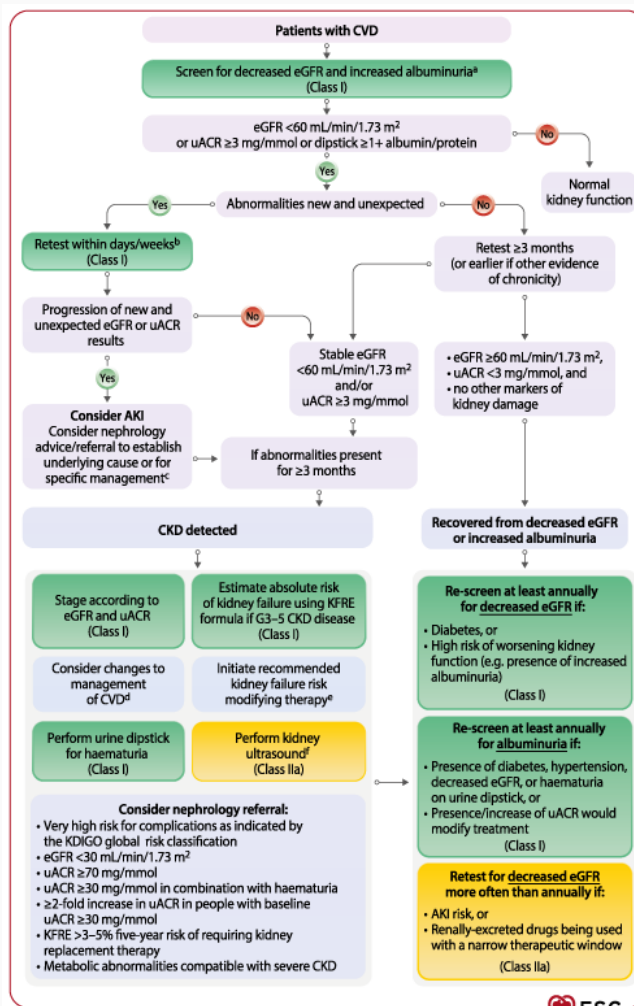
# Recommendations for screening for chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                                                            | Class | Level |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Screening for CKD using eGFR and increased albuminuria is recommended in patients with CVD to enable adequate CVD and CKD risk prediction, to guide use of CVD and CKD risk-modifying treatment, for drug dosing considerations, and to allow timely nephrology referral.                  | I     | C     |
| At least annual re-testing for eGFR is recommended in patients with diabetes or at high risk of worsening kidney function (e.g. presence of increased albuminuria or decreased eGFR) to detect CKD progression or re-classify CKD staging.                                                 | I     | C     |
| At least annual re-testing for albuminuria is recommended in patients with diabetes, hypertension, decreased eGFR, or haematuria on urine dipstick, or when presence or an increase in albuminuria would modify treatment, to re-classify CKD stage and optimize risk-modifying treatment. | I     | C     |
| Repeat measurement of eGFR and albuminuria within days to weeks is recommended in patients with newly detected unexpected decreased eGFR and/or increased albuminuria to allow a timely diagnosis of AKI for diagnostic evaluation and management.                                         | I     | C     |

## Recommendations for screening for chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                                                 | Class     | Level    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|
| Re-testing for decreased eGFR more often should be considered for patients at risk of AKI and for patients using renally excreted drugs with a narrow therapeutic window to allow timely identification of AKI or progressive CKD and for drug dose adjustment. | <b>Ia</b> | <b>C</b> |

# Figure 4 Screening algorithm for diagnosis and staging of chronic kidney disease in cardiovascular disease



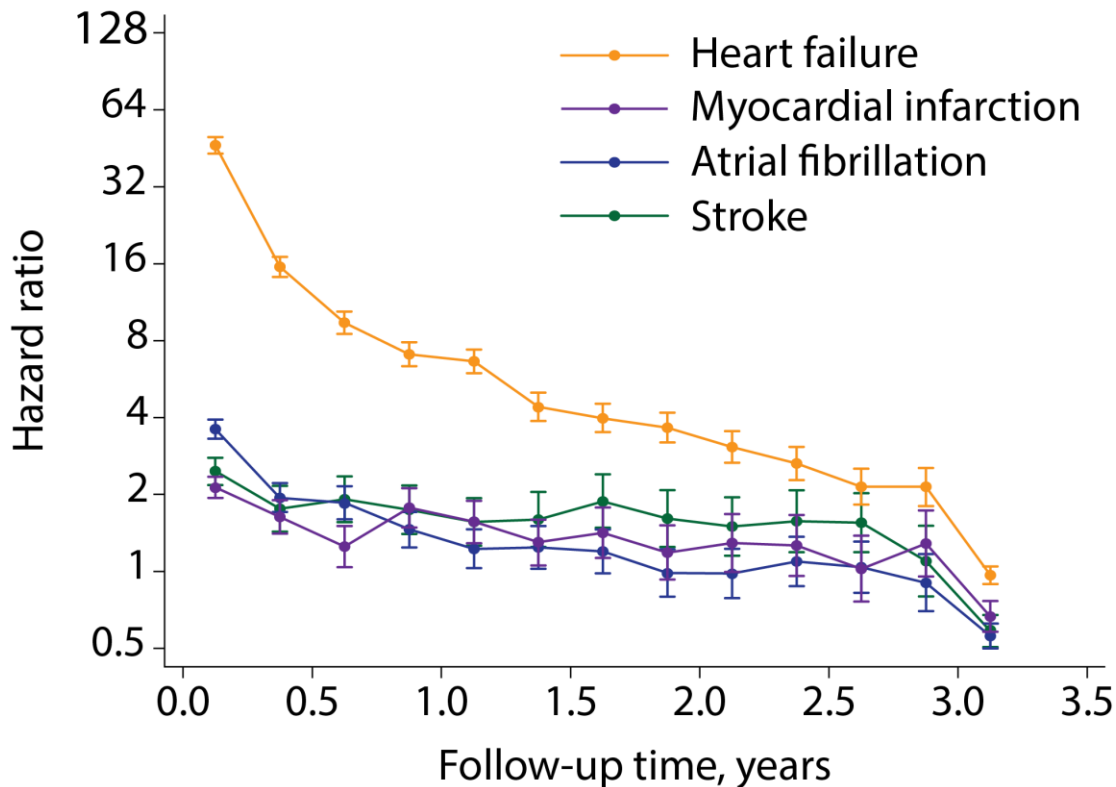
# Recommendations for drug dosing in chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                    | Class | Level |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Assessment of the benefits vs. potential harms when prescribing medications which are potentially nephrotoxic is recommended in patients with CKD with the aim of minimizing the risk of AKI or accelerating CKD progression.                                                                                                                                                                                                      | I     | C     |
| Monitoring eGFR, relevant electrolytes, QT interval on electrocardiography (when indicated), and therapeutic medication levels (where possible) is recommended in patients with CKD who are receiving medications which are renally excreted and have narrow therapeutic windows or potential adverse effects when eGFR is decreased, with the aim of preventing adverse drug reactions and ensuring safe and effective treatment. | I     | C     |

# Recommendations for assessing kidney failure risk in chronic kidney disease

| Recommendations                                                                                                                                                                                                                                           | Class | Level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Using a validated Kidney Failure Risk Equation (KFRE), is recommended in patients with CKD categories G3–5 when assessing an individual's absolute risk for progression to require KRT to allow timely referral for nephrology evaluation and management. | I     | A     |

**Figure 5 Risk of kidney replacement therapy by time since incident cardiovascular disease**



# Recommendation for cardiovascular disease assessment in chronic kidney disease

| Recommendations                                                                                                                                                                                                                                        | Class | Level |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Assessment for symptoms and signs of hypertension, ASCVD, HF, arrhythmias, and valvular heart disease is recommended for patients with CKD at each clinical encounter to help achieve early diagnosis and management of these cardiovascular diseases. | I     | C     |

# Recommendations for assessing cardiovascular risk in chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                                                                                                   | Class | Level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Where available, risk scores incorporating eGFR and albuminuria are recommended for cardiovascular risk assessment in patients with CKD to provide an accurate estimate.                                                                                                                                                          | I     | B     |
| Patients with severe CKD (eGFR <30 mL/min/1.73 m <sup>2</sup> with any level of albuminuria; eGFR 30–44 mL/min/1.73 m <sup>2</sup> and uACR 3–30 mg/mmol; uACR >30 mg/mmol with any level of eGFR) are recommended to be considered at relatively very high CVD risk to optimize implementation of primary prevention strategies. | I     | B     |
| Patients with moderate CKD (eGFR 30–44 mL/min/1.73 m <sup>2</sup> and uACR <3 mg/mmol; eGFR 45–59 mL/min/1.73 m <sup>2</sup> and uACR 3–30 mg/mmol) without other major risk factors are recommended to be considered at relatively high CVD risk to optimize implementation of primary prevention strategies.                    | I     | B     |

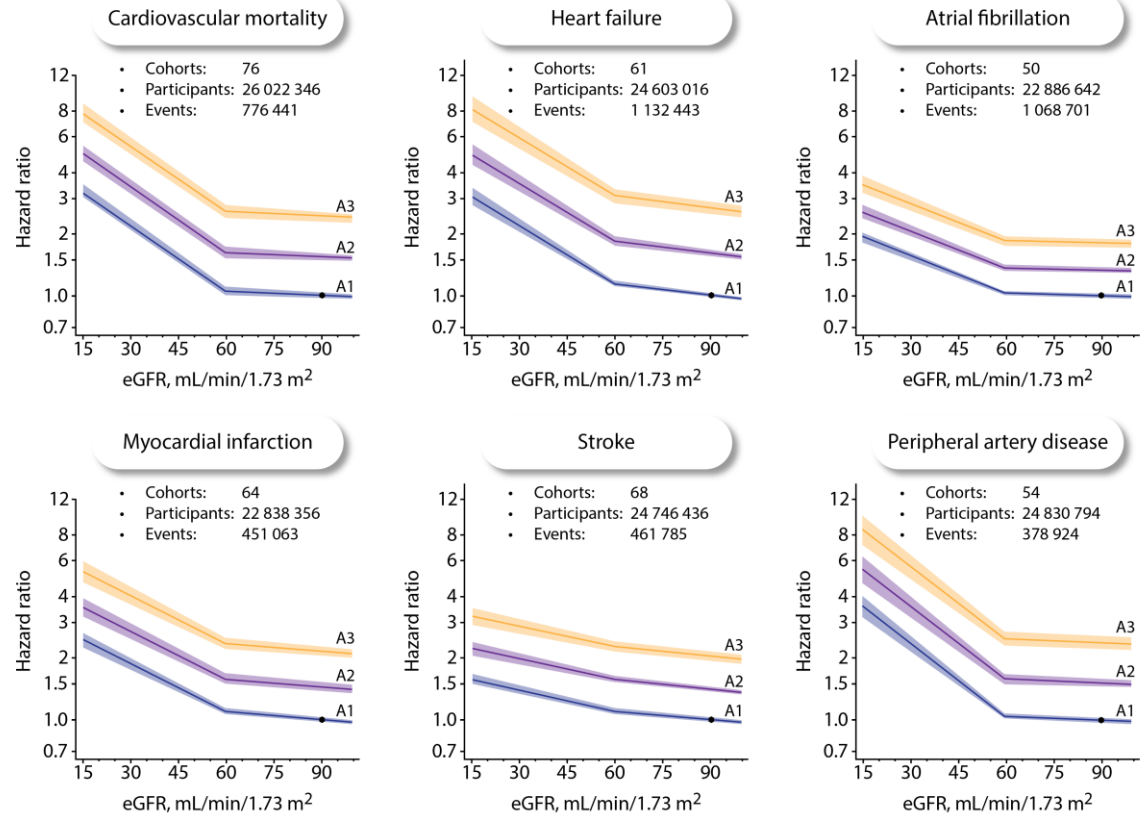
## Recommendations for assessing cardiovascular risk in chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                                                                                                    | Class | Level |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| When an accurate estimation of an individual's absolute risk is required, the use of the SCORE2 (aged <70 years) or SCORE2-OP (age ≥70 years) risk calculators with the 'CKD Add-on' is recommended in patients who have CKD without type 2 diabetes and without symptomatic ASCVD to estimate 10-year ASCVD risk. | I     | B     |
| When an accurate estimation of an individual's absolute risk is required, the use of the SCORE2-Diabetes risk score is recommended in patients with both CKD and type 2 diabetes but without symptomatic ASCVD to estimate 10-year ASCVD risk.                                                                     | I     | B     |
| When an accurate estimation of an individual's absolute risk is required, the use of the SMART2 risk score is recommended in patients who have CKD and pre-existing atherosclerotic vascular disease to estimate the risk of recurrent vascular events.                                                            | I     | B     |

# Scores for estimating risk of cardiovascular disease incorporating estimated glomerular filtration rate

| Risk score                             | Diabetes status         | Age criteria | Outcome                                                     |
|----------------------------------------|-------------------------|--------------|-------------------------------------------------------------|
| Without pre-existing ASCVD             |                         |              |                                                             |
| SCORE2 + CKD Add-On                    | Without type 2 diabetes | <70 years    | 10-year risk of ASCVD                                       |
| SCORE2-OP + CKD Add-On                 | Without type 2 diabetes | ≥70 years    | 10-year risk of ASCVD                                       |
| SCORE2-Diabetes                        | With type 2 diabetes    | ≥40 years    | 10-year risk of ASCVD                                       |
| SCORE2-HF                              | Any diabetes status     | ≥40 years    | Incident HF                                                 |
| With pre-existing ASCVD and without HF |                         |              |                                                             |
| SMART2                                 | Any diabetes status     | ≥40 years    | Recurrent vascular events                                   |
| SMART2-HF                              | Any diabetes status     | ≥40 years    | Incident HF                                                 |
| With pre-existing HFpEF                |                         |              |                                                             |
| LIFE-Preserved                         | Any diabetes status     | ≥40 years    | Any-year or lifetime risk of HF hospitalization or CV death |

# Figure 6 Associations between estimated glomerular filtration rate and albuminuria categories with subsequent risk of cardiovascular disease



# Recommendations for lifestyle factors in chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                                                                                               | Class | Level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Smoking cessation, supportive care, and referral to providers and programmes for smoking cessation are recommended in patients with CKD who smoke tobacco, to reduce the risk of cardiovascular events and mortality.                                                                                                                         | I     | B1    |
| An individualized diet with low sodium intake (<2 g sodium per day, equivalent to <5 g sodium chloride), and high intake of vegetables is recommended in patients with CKD, to reduce blood pressure, albuminuria, and cardiovascular events, and slow CKD progression.                                                                       | I     | B1    |
| Weight loss through individualized lifestyle modification with or without weight loss medication is recommended in patients with CKD who are overweight or obese (BMI $\geq 25$ kg/m <sup>2</sup> ) to achieve a BMI 18.5–25 kg/m <sup>2</sup> to reduce blood pressure, improve lipid profile, and reduce risk of worsening of CKD category. | I     | B2    |
| Physical activity (e.g. $\geq 150$ minutes of moderate-intensity physical activity throughout the week, where possible) is recommended in patients with CKD to reduce blood pressure and cardiovascular risk.                                                                                                                                 | I     | B2    |

# Recommendations for blood pressure management in chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                      | Class | Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Screening by attended standardized office BP measurement is recommended in all patients with CKD for early identification and confirmation of elevated BP/hypertension.                                                                                              | I     | A     |
| Intensified BP control by lifestyle optimization and BP-lowering medications is recommended in patients with CKD and $\text{eGFR} \geq 30 \text{ mL/min/1.73 m}^2$ , who have confirmed standardized office BP $\geq 130/80 \text{ mmHg}$ to reduce the risk of CVD. | I     | A     |
| A systolic BP target of 120–129 mmHg, if tolerated, is recommended in patients with CKD and $\text{eGFR} \geq 30 \text{ mL/min/1.73 m}^2$ to reduce the risk of CVD and reduce albuminuria.                                                                          | I     | A     |
| Performing out-of-office (i.e. ABPM or HBPM) is recommended in patients with CKD to identify white-coat hypertension, masked hypertension, or elevated night-time BP.                                                                                                | I     | C     |
| Individualized BP targets are recommended in patients with $\text{eGFR} < 30 \text{ mL/min/1.73 m}^2$ to reduce risk of CVD.                                                                                                                                         | I     | C     |

# Recommendations for the management of dyslipidaemia in patients with chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                              | Class | Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| <b>Lipid evaluation</b>                                                                                                                                                                                                                      |       |       |
| Assessment of a baseline extended lipid panel, including total cholesterol, LDL-C, HDL-C, and triglycerides is recommended in patients with newly diagnosed CKD to identify lipid abnormalities and support cardiovascular risk prediction.  | I     | C     |
| Assessment of non-HDL cholesterol (or apolipoprotein B, if available) should be considered in people with CKD for cardiovascular risk prediction and in those with high triglycerides or diabetes as an alternative treatment goal to LDL-C. | IIa   | C     |
| <b>CKD without KRT</b>                                                                                                                                                                                                                       |       |       |
| Irrespective of lipid levels, an intensive statin-based regimen shown to be safe in CKDc is recommended in patients with CKD with an eGFR <60 mL/min/1.73 m <sup>2</sup> not on KRT to reduce risk of major atherosclerotic events.          | I     | A     |
| <b>Kidney transplant recipients</b>                                                                                                                                                                                                          |       |       |
| A statin-based regimen should be considered in kidney transplant recipients to reduce the risk of major atherosclerotic events.                                                                                                              | IIa   | B1    |

# Recommendations for the management of dyslipidaemia in patients with chronic kidney disease (2)

| Recommendations                                                                                                                                                          | Class      | Level    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| CKD treated by dialysis                                                                                                                                                  |            |          |
| Continuation of a statin-based regimen may be considered on dialysis initiation to ensure ongoing cardiovascular protection.                                             | <b>IIb</b> | <b>C</b> |
| Initiating a statin-based regimen may be considered in patients on dialysis with a history of ASCVD to reduce the risk of recurrent atherosclerotic events.              | <b>IIb</b> | <b>C</b> |
| A statin-based regimen may be considered in patients on dialysis in whom kidney transplantation is planned to reduce the long-term risk of major atherosclerotic events. | <b>IIb</b> | <b>C</b> |

# Recommendations for the use of antiplatelet therapy in chronic kidney disease without symptomatic atherosclerotic cardiovascular disease

| Recommendations                                                                                                                                                                                                                                      | Class      | Level     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Routine use of antiplatelet therapy is not recommended in patients with CKD with eGFR <60 mL/min/1.73 m <sup>2</sup> without symptomatic ASCVD because increased risk of major bleeding may outweigh any reduction in risk of cardiovascular events. | <b>III</b> | <b>B1</b> |

# Recommendations for specific pharmacological interventions to reduce risk of kidney failure and/or cardiovascular disease in chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                            | Class | Level |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Treatment with maximally tolerated ACEI or ARB is recommended in patients with CKDc to achieve target blood pressure, reduce risk of progression of CKD and risk of cardiovascular events.                                                                 | I     | A     |
| An SGLT2 inhibitor is recommended in patients with CKD who have type 2 diabetes with eGFR $\geq 20$ mL/min/1.73 m <sup>2</sup> to reduce risk of kidney failure and risk of cardiovascular events, irrespective of glycaemic control.                      | I     | A     |
| An SGLT2 inhibitor is recommended in patients with CKD without diabetes with eGFR $\geq 20$ mL/min/1.73 m <sup>2</sup> and uACR $\geq 20$ mg/mmol ( $\geq 200$ mg/g) to reduce the risk of kidney failure and cardiovascular events.                       | I     | A     |
| A non-steroidal MRA (finerenone) is recommended in patients with CKD who have type 2 diabetes with eGFR $\geq 25$ mL/min/1.73 m <sup>2</sup> and uACR $\geq 3$ mg/mmol ( $\geq 30$ mg/g) to reduce the risk of kidney failure and cardiovascular events.d, | I     | A     |

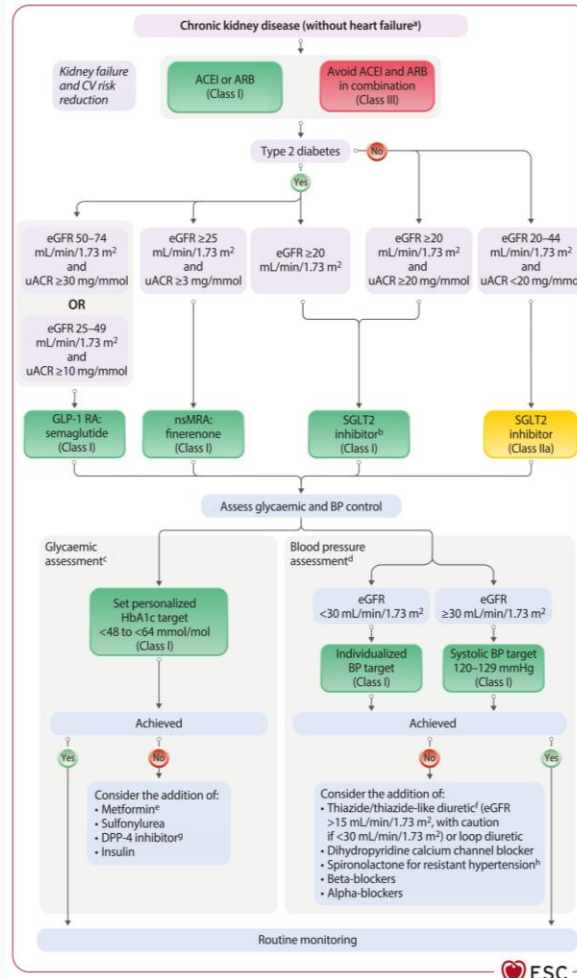
## Recommendations for specific pharmacological interventions to reduce risk of kidney failure and/or cardiovascular disease in chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                                                                                                                                                                              | Class | Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| A GLP-1RA (subcutaneous semaglutide 1.0 mg/week after dose titration) is recommended in patients with CKD who have type 2 diabetes with eGFR 25–49 mL/min/1.73 m <sup>2</sup> and uACR ≥10 mg/mmol (≥100 mg/g) or eGFR 50–74 mL/min/1.73 m <sup>2</sup> and uACR ≥30 mg/mmol (≥300 mg/g) to reduce the risk of CKD progression and cardiovascular events, irrespective of glycaemic control. | I     | A     |
| An SGLT2 inhibitor should be considered in patients with CKD without diabetes with eGFR 20–44 mL/min/1.73 m <sup>2</sup> and uACR <20 mg/mmol to slow kidney disease progression.                                                                                                                                                                                                            | IIa   | B1    |
| Combined use of an ARB with an ACEI is not recommended in patients with CKD as the risk of AKI and severe hyperkalaemia outweighs any added cardiorenal benefits.                                                                                                                                                                                                                            | III   | A     |

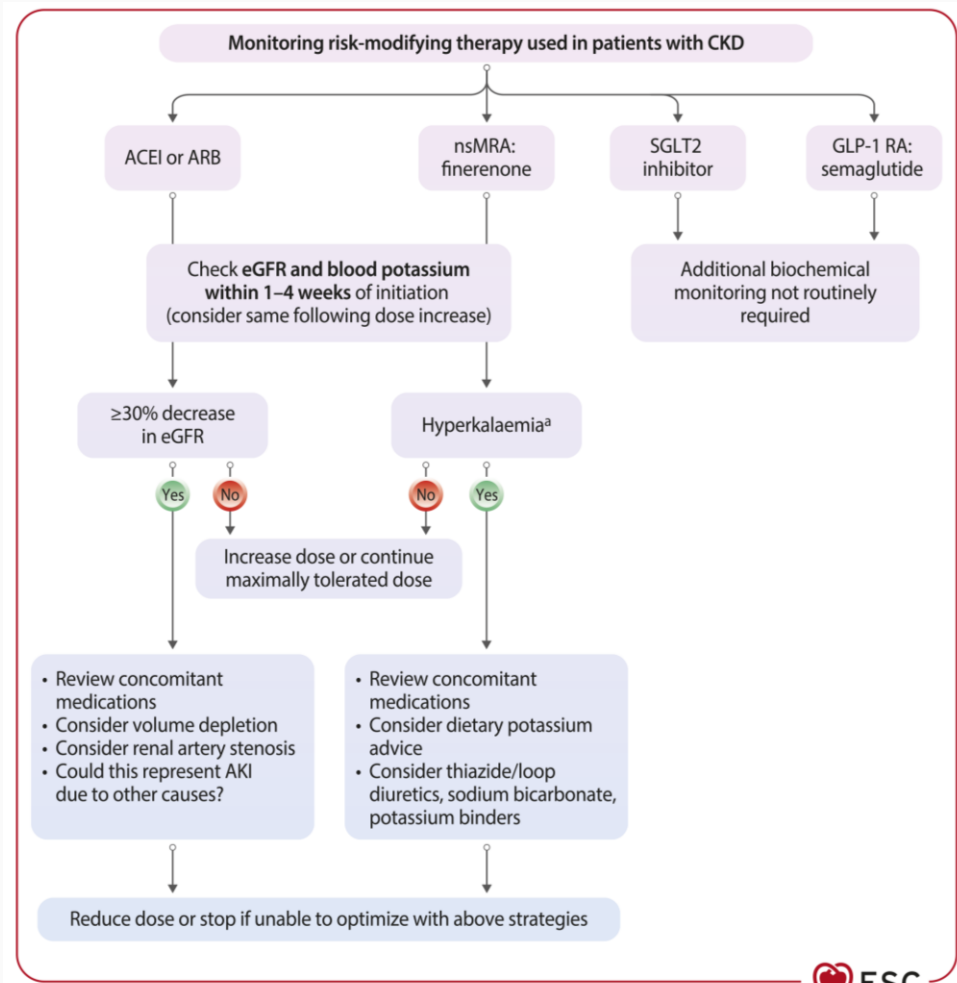
# Recommendations for glucose lowering in patients with diabetes and chronic kidney disease

| Recommendations                                                                                                                                                                                                                                    | Class | Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| SGLT2 inhibitors and GLP-1RAs are recommended as first-line agents in patients with type 2 diabetes and CKD to lower blood glucose, due to their benefits on CKD progression and risk of cardiovascular events.                                    | I     | A     |
| Insulin is recommended in patients with type 2 diabetes and CKD if blood glucose targets are not reached with oral agents or GLP-1RA therapy, for additional blood glucose lowering.                                                               | I     | A     |
| Personalized HbA1c targets within the range of <48 mmol/mol (<6.5%) to <64 mmol/mol (<8.0%) in patients with CKD who have diabetes are recommended to reduce microvascular complications, with a target of <53 mmol/mol (<7.0%) wherever possible. | I     | B1    |
| Metformin should be considered in patients with CKD with eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> and type 2 diabetes for additional blood glucose lowering.                                                                                      | IIa   | B1    |
| Sulfonylurea therapy may be considered in patients with CKD and type 2 diabetes for additional blood glucose lowering.                                                                                                                             | IIb   | C     |
| A DPP-4 inhibitor may be considered in patients with CKD and type 2 diabetes if GLP-1RA therapy is not tolerated or contraindicated, for additional blood glucose lowering.                                                                        | IIb   | B1    |

# Figure 7 Algorithm for use of risk-modifying therapy and glucose lowering in patients with chronic kidney disease



## Figure 8 Simple guide to clinical and biochemical monitoring of risk-modifying therapy used in patients with chronic kidney disease



# Recommendations for heart failure screening and diagnosis in patients with chronic kidney disease

| Recommendations                                                                                                                                                                | Class      | Level    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Screening for clinical suspicion of HF                                                                                                                                         |            |          |
| Assessment of natriuretic peptides may be considered in patients with CKD as a screening tool for structural heart disease, cardiac dysfunction, or prediction of incident HF. | <b>IIb</b> | <b>C</b> |

# Recommendations for treatment of chronic heart failure in patients with chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                       | Class | Level |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Treatment irrespective of LVEF                                                                                                                                                                                                                        |       |       |
| Treatment with loop diuretics is recommended in patients with HF and CKD to alleviate symptoms/signs of fluid overload and improve exercise capacity.                                                                                                 | I     | A     |
| Treatment with an SGLT2 inhibitor is recommended in patients with HF and CKD with eGFR $\geq 20$ mL/min/1.73 m <sup>2</sup> irrespective of LVEF, to reduce the risk of (repeated) HF hospitalization and cardiovascular death.                       | I     | A     |
| Treatment with a steroidal MRA is recommended in patients with HF and CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> and HFrEF, to reduce the risk of HF hospitalization and death.                                                            | I     | A     |
| Treatment with a MRA, either steroidal or non-steroidal, should be considered in patients with HF with LVEF $\geq 40\%$ and CKD with an eGFRc $\geq 25$ mL/min/1.73 m <sup>2</sup> to reduce the risk of HF hospitalization and cardiovascular death. | IIa   | B1    |

## Recommendations for treatment of chronic heart failure in patients with chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                                        | Class | Level |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| <b>HFrEF</b>                                                                                                                                                                                                                                           |       |       |
| Treatment with either ACEI, ARNI, or ARB (if ACEI or ARNI not tolerated) is recommended in patients with HFrEF and CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> to reduce the risk of HF hospitalization and death.                           | I     | A     |
| Treatment with an ACEI is recommended in patients with CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> and asymptomatic left ventricular dysfunction (LVEF $\leq 35\%$ ) to reduce the risk of incident HF and HF-related hospitalizations.      | I     | B1    |
| Treatment with a beta-blocker is recommended in patients with HFrEF and CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> to reduce the risk of HF hospitalization and death.                                                                      | I     | A     |
| Treatment with an ARNI (sacubitril/valsartan) should be considered in patients with HFrEF and CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> as an alternative to ACEI/ARB to reduce the risk of CKD progression (or slow the decline in eGFR). | IIa   | B1    |

## Recommendations for treatment of chronic heart failure in patients with chronic kidney disease (3)

| Recommendations                                                                                                                                                                                                                                                                                                                                                           | Class      | Level    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| HFrEF (continued)                                                                                                                                                                                                                                                                                                                                                         |            |          |
| Treatment with a combination of hydralazine-isosorbide dinitrate should be considered in self-identified Black patients with HFrEF and CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> and dilated left ventricle in combination with LVEF <45%, or LVEF $\leq 35\%$ , when added to conventional HFrEF therapy to reduce the risk of HF hospitalization and death. | <b>IIa</b> | <b>C</b> |
| Treatment with an ACEI or ARB may be considered in patients with HFrEF and CKD with an eGFR 15–29 mL/min/1.73 m <sup>2</sup> to reduce the risk of HF hospitalization and cardiovascular death.                                                                                                                                                                           | <b>IIb</b> | <b>C</b> |
| Treatment with a beta-blocker may be considered in patients with HFrEF and CKD with an eGFR 15–29 mL/min/1.73 m <sup>2</sup> to reduce the risk of HF hospitalization and death.                                                                                                                                                                                          | <b>IIb</b> | <b>C</b> |

# Recommendations for treatment of chronic heart failure in patients with chronic kidney disease (3)

| Recommendations                                                                                                                                                                                                                                                                                                      | Class | Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Additional medical therapy in HFrEF                                                                                                                                                                                                                                                                                  |       |       |
| Additional medical therapy with cardiac glycosides (digoxin or digitoxin) should be considered in patients with HFrEF and CKD with eGFR >20 mL/min/1.73 m <sup>2</sup> to reduce the risk of HF hospitalization and death.                                                                                           | Ila   | B1    |
| Additional medical therapy with vericiguat may be considered in patients with HFrEF and CKD with eGFR ≥30 mL/min/1.73 m <sup>2</sup> to reduce the risk of HF hospitalization and cardiovascular death.                                                                                                              | IIb   | B1    |
| Additional medical therapy with ivabradine should be considered in patients with HFrEF and CKD with eGFR ≥20 mL/min/1.73 m <sup>2</sup> to reduce the risk of cardiovascular and HF-related events.                                                                                                                  | Ila   | B1    |
| HFpEF                                                                                                                                                                                                                                                                                                                |       |       |
| Treatment with a GLP-1RA (semaglutide) or glucose-dependent insulinotropic polypeptide and GLP-1RA (tirzepatide) should be considered in patients with HFpEF and CKD with an eGFR ≥15 mL/min/1.73 m <sup>2</sup> and LVEF ≥45% with or without diabetes who are obese, to reduce weight and improve quality of life. | Ila   | B1    |

# Recommendations for management of comorbidities in patients with heart failure and chronic kidney disease

| Recommendations                                                                                                                                                    | Class      | Level     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Iron deficiency                                                                                                                                                    |            |           |
| Treatment with intravenous iron supplementation should be considered in patients with HFrEF and CKD with iron deficiency to reduce the risk of HF hospitalization. | <b>Ila</b> | <b>B1</b> |

# Management of patients with decompensated heart failure and chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                                         | Class | Level |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Initiation of an SGLT2 inhibitor is recommended early during admission in patients with DHF and CKD with an eGFR $>20$ mL/min/ $1.73$ m <sup>2</sup> to improve signs and symptoms of congestion and reduce the risk of HF re-hospitalization.                          | I     | B1    |
| An initial intravenous diuretic dose $>40$ mg of furosemide-equivalent should be considered in patients with DHF with fluid overload and CKD (not on KRT) who are diuretic-naïve to achieve sufficient diuretic response.                                               | IIa   | B2    |
| Upfront treatment with acetazolamide in addition to adequately dosed intravenous loop diuretic therapy should be considered in patients with DHF and CKD with an eGFR $\geq 20$ mL/min/ $1.73$ m <sup>2</sup> on chronic loop diuretic therapy to improve decongestion. | IIa   | B1    |
| Assessment of diuretic response should be considered in patients with DHF and CKD (not on KRT) in the first days of hospitalization to optimize diuretic treatment and improve diuretic response.                                                                       | IIa   | B1    |

# Management of patients with decompensated heart failure and chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                                          | Class      | Level     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Upfront treatment with thiazides in addition to adequately dosed intravenous loop diuretic therapy may be considered in patients with DHF and CKD (not on KRT) to improve decongestion.                                                                  | <b>IIb</b> | <b>B1</b> |
| An initial intravenous diuretic dose equivalent to 2.5-times the daily home loop diuretic dose may be considered in patients with DHF and CKD (not on KRT) on chronic loop diuretic therapy to improve dyspnoea relief, weight loss, and net fluid loss. | <b>IIb</b> | <b>C</b>  |

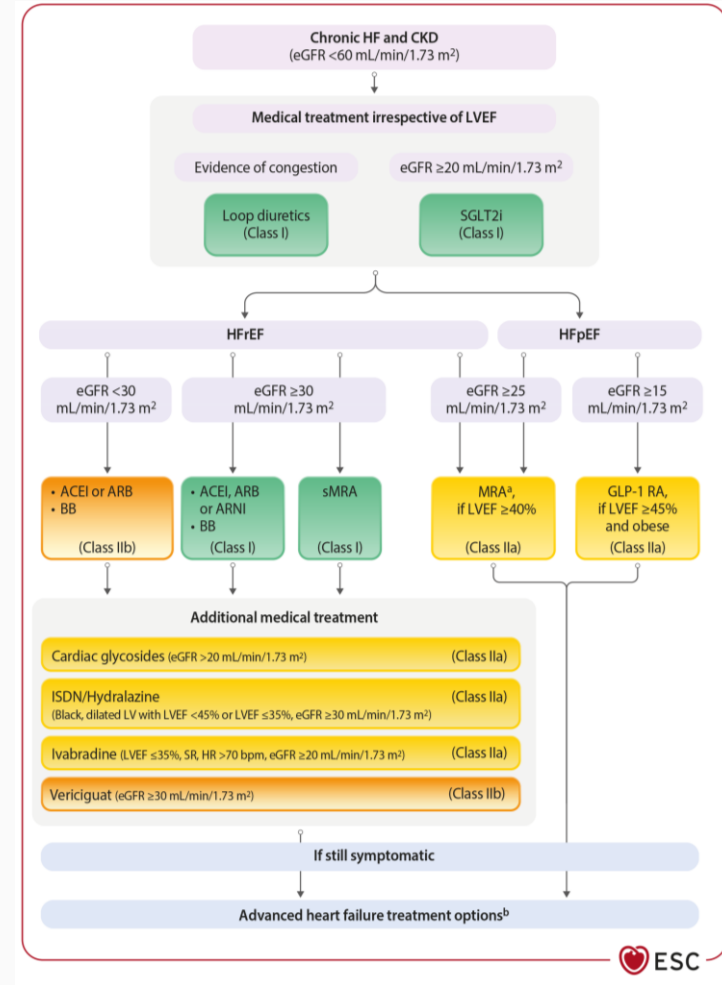
# Recommendations for management of heart failure therapy in the context of a drop in estimated glomerular filtration rate

| Recommendation                                                                                                                                                                                                                                       | Class      | Level    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Chronic heart failure                                                                                                                                                                                                                                |            |          |
| Continuation of a SGLT2 inhibitor should be considered in patients with HF and CKD in whom the eGFR progresses to $<20$ mL/min/1.73 m <sup>2</sup> , to avoid worsening of HF.                                                                       | <b>IIa</b> | <b>C</b> |
| Continuation of an ARNI should be considered in patients with HF and CKD in whom the eGFR progresses to $<30$ mL/min/1.73 m <sup>2</sup> , to avoid worsening of HF.                                                                                 | <b>IIa</b> | <b>C</b> |
| Decompensated heart failure                                                                                                                                                                                                                          |            |          |
| Continuation of decongestive therapy is recommended in congested patients with decompensated HF and CKD (not on KRT) who exhibit good diuretic response despite a rise in creatinine (up to 50%) to help achieve and maintain complete decongestion. | <b>I</b>   | <b>C</b> |

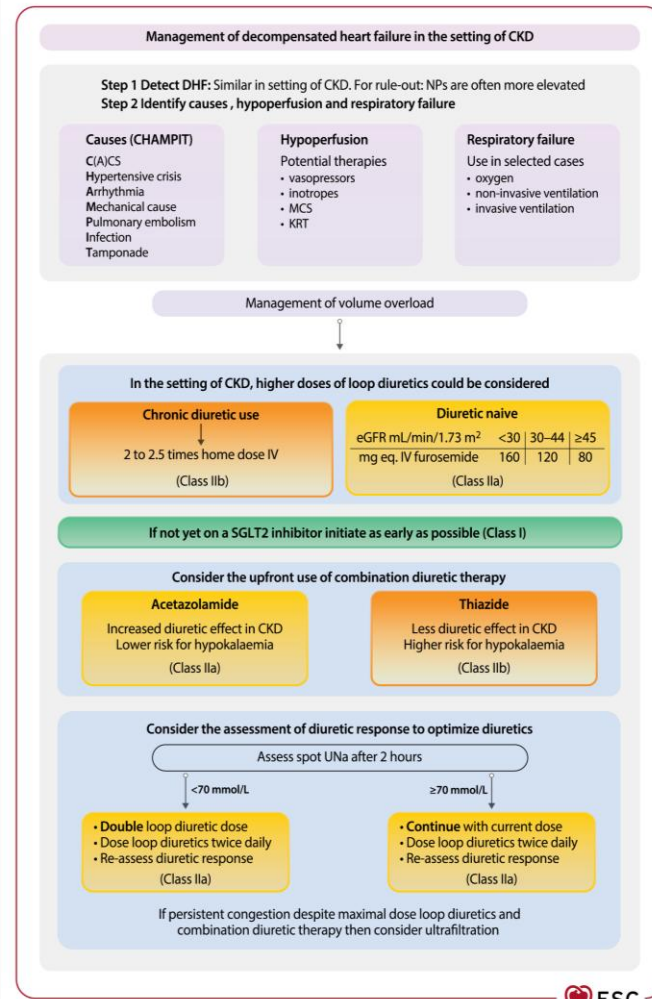
# Management of patients with advanced heart failure and chronic kidney disease

| Recommendation                                                                                                                                                                                                                                                                           | Class      | Level    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Evaluation by a heart–kidney transplant team should be considered in all patients with advanced HF eligible for heart transplant and a baseline eGFR <30 mL/min/1.73 m <sup>2</sup> to assess the need for dual organ transplant.                                                        | <b>IIa</b> | <b>C</b> |
| A strategy of haemodynamic optimization (tailored approach of applying diuretics, inotropes, and short-term MCS based on invasive haemodynamics) may be considered in patients with CKD (not on long-term KRT) and advanced HF to assess eligibility for durable MCS or transplantation. | <b>IIb</b> | <b>C</b> |
| Durable MCS is not recommended in patients with advanced HF with a baseline eGFR <30 mL/min/1.73 m <sup>2</sup> (which is unlikely to improve) who are not potential candidates for dual organ transplantation, to reduce the risk of death in this population.                          | <b>III</b> | <b>C</b> |

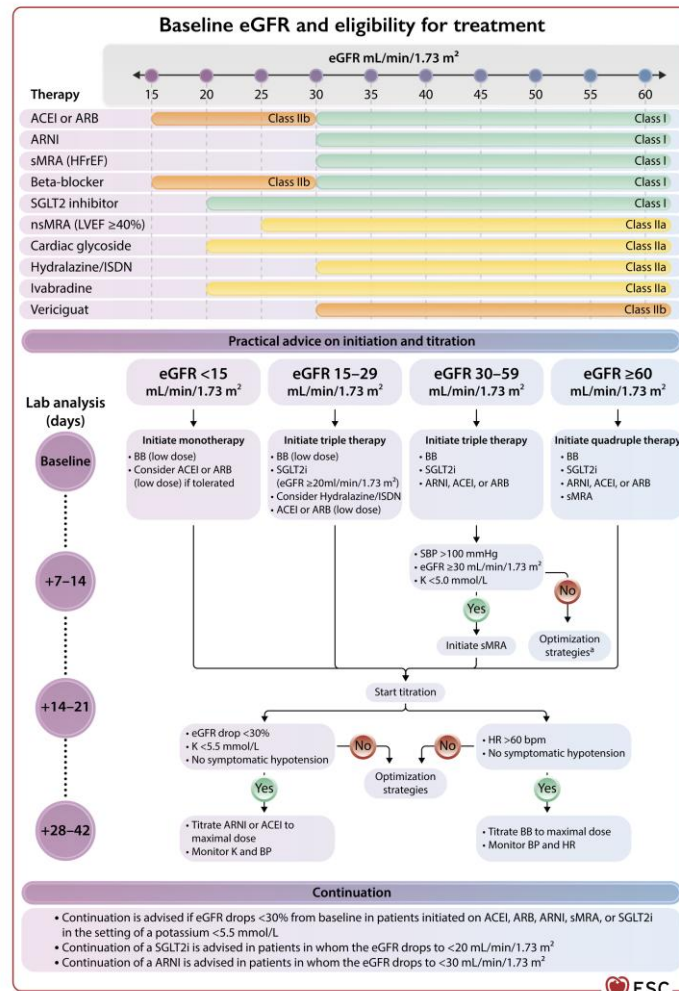
# Figure 9 Pharmacological treatment of patients with heart failure and chronic kidney disease



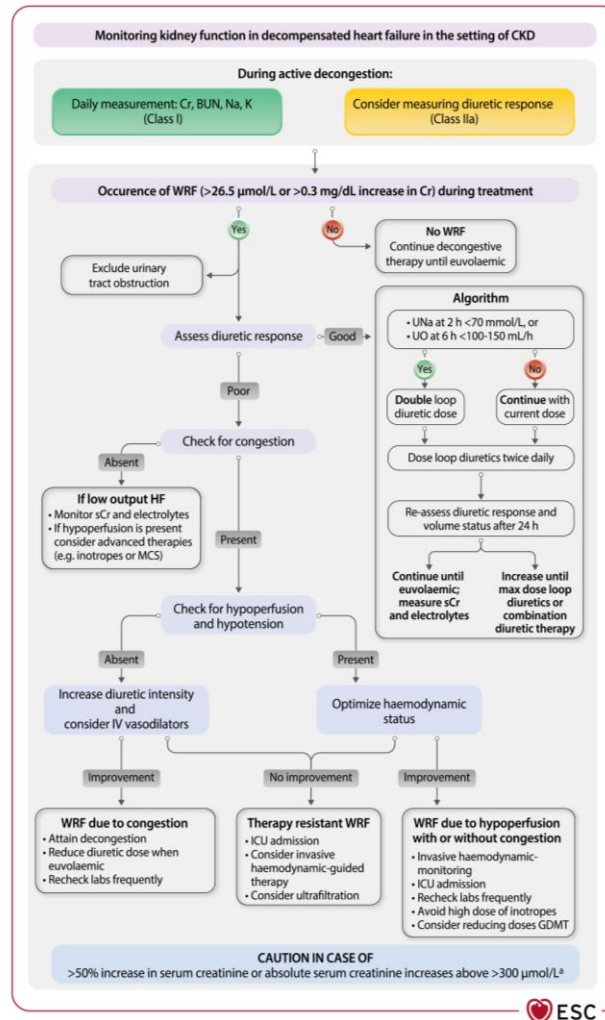
# Figure 10 Management of decompensated heart failure in chronic kidney disease



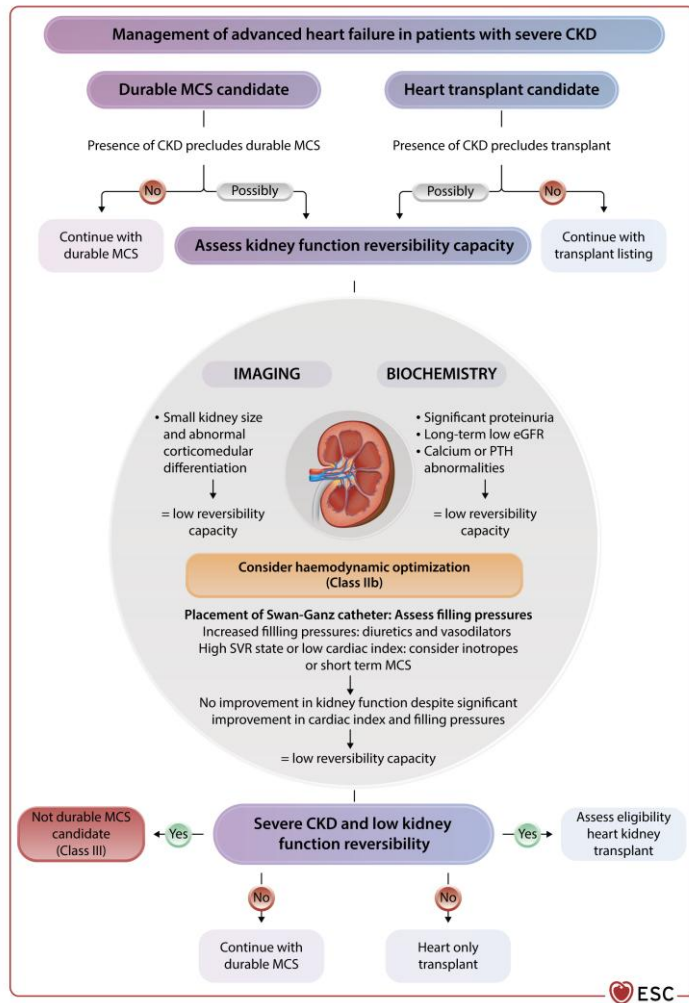
# Figure 11 Monitoring kidney function in the treatment of chronic heart failure with reduced ejection fraction and chronic kidney disease



# Figure 12 Monitoring kidney function in the treatment of decompensated heart failure and chronic kidney disease



# Figure 13 Advanced heart failure and chronic kidney disease assessment



# Recommendations for management of chronic coronary syndromes in chronic kidney disease (1)

| Recommendation                                                                                                                                                                                                                                                                                       | Class | Level |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| An individualized approach to diagnostic imaging is recommended in symptomatic patients with CKD, to detect obstructive CAD, improve risk stratification, and triage patients for intervention, balancing the relative benefits and potential risks associated with different diagnostic modalities. | I     | C     |
| Antiplatelet therapy with low-dose aspirin or clopidogrel is recommended in patients with CCS and CKD, without an indication for anticoagulation, to reduce the risk of major adverse cardiovascular events.                                                                                         | I     | C     |
| Antiplatelet therapy with clopidogrel should be considered in preference to low-dose aspirin in patients with CCS and CKD with eGFR >30 mL/min/1.73 m <sup>2</sup> , without an indication for anticoagulation, to reduce the risk of major adverse cardiovascular events.                           | IIa   | B1    |
| A statin-based regimen with or without ezetimibe is recommended in patients with CCS and CKD not on dialysis, to reduce the risk of major atherosclerotic events.                                                                                                                                    | I     | A     |
| Treatment with a beta-blocker, calcium channel blocker, long-acting nitrate, or a combination of these, is recommended in patients with CCS and CKD for the relief of angina or equivalent symptoms.                                                                                                 | I     | C     |

## Recommendations for management of chronic coronary syndromes in chronic kidney disease (2)

| Recommendation                                                                                                                                                                                                                                                                                                                                                                                                                         | Class | Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| An initial conservative strategy with optimal medical therapy is recommended in patients with CKD and CCS to reduce the risk of death or non-fatal MI.                                                                                                                                                                                                                                                                                 | I     | B1    |
| Individualized decision-making regarding revascularization (PCI or CABG vs. optimal medical therapy) and type of revascularization (PCI vs. CABG) is recommended in patients with CCS and CKD, following multidisciplinary team decision-making, consideration of anatomical suitability, short- and long-term risks, technical limitations and benefits of each intervention, and patient preferences, to optimize clinical outcomes. | I     | C     |

## Recommendations for management of chronic coronary syndromes in chronic kidney disease (3)

| Recommendation                                                                                                                                                                                                                                                                                                                                                                                      | Class      | Level     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Treatment with a PCSK9 inhibitor as add-on therapy should be considered in patients with CCS and CKD (eGFR $\geq 20$ mL/min/1.73 m <sup>2</sup> ) with LDL-C $\geq 70$ mg/dL ( $\geq 1.8$ mmol/L) or non-HDL-C $\geq 100$ mg/dL ( $\geq 2.6$ mmol/L) despite moderate- or high-intensity statin-based regimen with or without ezetimibe, to reduce the risk of major adverse cardiovascular events. | <b>IIa</b> | <b>B1</b> |
| Ranolazine should be considered for patients with CCS and CKD with eGFR $>30$ mL/min/1.73 m <sup>2</sup> , who remain symptomatic despite treatment with beta-blockers/calcium channel blockers/long-acting nitrates (especially if patients have a slow heart rate and low systolic blood pressure) for relief of angina or equivalent symptoms.                                                   | <b>IIa</b> | <b>C</b>  |
| Abbreviated DAPT duration (1-3 months) should be considered for patients with CKD and CCS undergoing elective PCI, without an indication for anticoagulation, to reduce the risk of bleeding.                                                                                                                                                                                                       | <b>IIa</b> | <b>B2</b> |
| A radial approach should be considered over a femoral approach, when feasible, in patients with CKD (including those on KRT) undergoing PCI for CCS, to reduce the risk of bleeding and in-hospital mortality.                                                                                                                                                                                      | <b>IIa</b> | <b>C</b>  |

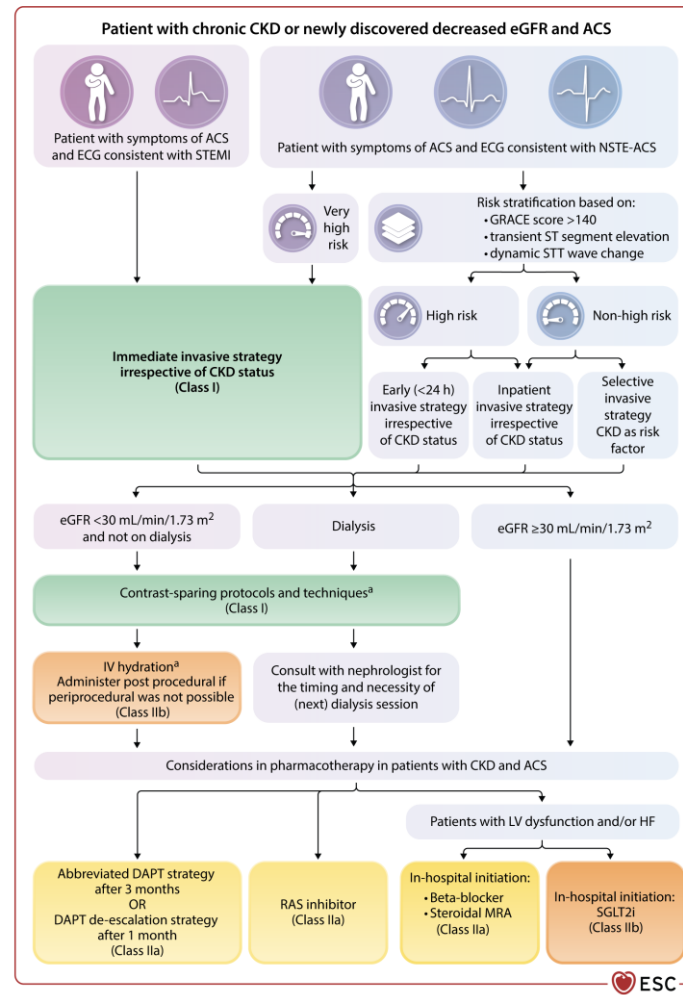
## Recommendations for management of chronic coronary syndromes in chronic kidney disease (4)

| Recommendation                                                                                                                                                                                                                                                                                                                               | Class      | Level     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Treatment with icosapent ethyl as add-on therapy to a statin may be considered in patients with CCS and CKD (eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> ) with a fasting triglyceride level $>1.5$ mmol/L ( $>135$ mg/dL) and LDL-C $>1.04$ mmol/L ( $>40$ mg/dL) to reduce the risk of major adverse cardiovascular events.                  | <b>IIb</b> | <b>B1</b> |
| Treatment with a statin-based regimen with or without ezetimibe may be considered in patients with CCS and CKD on dialysis, to reduce the risk of recurrent atherosclerotic events.                                                                                                                                                          | <b>IIb</b> | <b>C</b>  |
| CABG may be considered in preference to PCI in patients with CKD and complex CAD (multi-vessel and/or left main disease), considering anatomical suitability, short- and long-term risks, technical limitations and benefits of each intervention, and patient preference, to reduce the long-term risks of MI and repeat revascularization. | <b>IIb</b> | <b>B2</b> |

## Recommendations for management of chronic coronary syndromes in chronic kidney disease (5)

| Recommendation                                                                                                                                                                                                                                                                                            | Class      | Level    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Ivabradine may be considered in patients with CCS and CKD with eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> , to reduce angina frequency and improve exercise capacity in the setting of reduced left ventricular systolic function (LVEF <40%), especially if patients have a resting heart rate >70 b.p.m. | <b>IIb</b> | <b>C</b> |
| Trimetazidine may be considered in patients with CCS and CKD with eGFR >30 mL/min/1.73 m <sup>2</sup> , in combination with beta-blockers and/or calcium channel blockers, in patients who remain symptomatic, for the relief of angina or equivalent symptoms.                                           | <b>IIb</b> | <b>C</b> |
| Nicorandil may be considered in patients with CCS and CKD with any eGFR, including those on KRT, in combination with beta-blockers and/or calcium channel blockers, for the relief of angina or equivalent symptoms in patients who remain symptomatic.                                                   | <b>IIb</b> | <b>C</b> |

# Figure 14 Algorithm for management of acute coronary syndrome in chronic kidney disease



# Recommendations for acute coronary syndrome in chronic kidney disease

| Recommendation                                                                                                                                                                                                                      | Class | Level |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Avoiding delays in immediate or early invasive strategies is recommended in patients with CKD with STEMI or NSTEMI-ACS with high or very-high risk to ensure timely treatment.                                                      | I     | C     |
| Early in-hospital treatment with an ACEI should be considered in patients with ACS and CKD with eGFR >20 mL/min/1.73 m <sup>2</sup> to reduce the risk of all-cause mortality.                                                      | IIa   | C     |
| In-hospital initiation of beta-blocker and steroidal MRA treatment should be considered in patients with ACS and CKDc complicated by left ventricular dysfunction or HF to reduce the risk of cardiovascular events.                | IIa   | B1    |
| Abbreviated DAPT duration (3 months) or DAPT de-escalation after 1–3 months should be considered in patients with ACS and CKD without an indication for anticoagulation to reduce the risk of major bleeding.                       | IIa   | B1    |
| In-hospital initiation of an SGLT2 inhibitor in patients with ACS and CKD with an eGFR >20 mL/min/1.73 m <sup>2</sup> complicated by left ventricular dysfunction or HF may be considered to reduce the risk of HF hospitalization. | IIb   | C     |

# Recommendations for diagnosis and management of peripheral artery disease in chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                                                                                   | Class | Level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| The addition of toe pressures and/or toe–brachial indices when testing for PAD is recommended in patients with CKD to diagnose the underlying disease and direct management.                                                                                                                                                      | I     | C     |
| An individualized multidisciplinary treatment approach should be considered in patients with CKD and complicated chronic limb-threatening ischaemia to ensure CKD-related considerations are included in decision-making and to ensure patients with complicated comorbidity can benefit from revascularization, where indicated. | Ila   | C     |
| Screening for CVD in other vascular territories should be considered in patients with CKD undergoing aortic surgery or PAD revascularization to inform peri-procedural risk stratification and a holistic approach to management of CVD.                                                                                          | Ila   | C     |

# Recommendations for atherothrombotic stroke prevention and treatment in patients with cerebrovascular disease or other ASCVD, and chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                                                          | Class | Level |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Antiplatelet agents                                                                                                                                                                                                                                                                      |       |       |
| Antiplatelet therapy is recommended in patients with CKD (including those on KRT) with a history of ischaemic stroke to reduce the risk of recurrent stroke.                                                                                                                             | I     | B2    |
| The addition of anticoagulation with very low-dose rivaroxaban (2.5 mg b.i.d.) to low-dose aspirin (75–100 mg o.d.) may be considered in patients with CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> , established ASCVD, and low risk of bleeding to reduce the risk of stroke. | IIb   | B1    |

# Recommendations for atherothrombotic stroke prevention and treatment in patients with cerebrovascular disease or other ASCVD, and chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                                                                                                                                                             | Class | Level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Lipid-lowering therapy                                                                                                                                                                                                                                                                                                                                                      |       |       |
| LDL-C lowering with statin-based therapy or a statin/ezetimibe combination regimen is recommended in patients with CKD not on dialysis and a history of ischaemic stroke to reduce the risk of recurrent stroke.                                                                                                                                                            | I     | A     |
| Treatment with a PCSK9 inhibitor as add-on therapy may be considered in patients with a history of ischaemic stroke and CKD with an eGFR $\geq 20$ mL/min/1.73 m <sup>2</sup> and LDL-C $\geq 70$ mg/dL ( $\geq 1.8$ mmol/L) or non-HDL-C $\geq 100$ mg/dL ( $\geq 2.6$ mmol/L) despite moderate- or high-intensity statin regimen, to reduce the risk of recurrent stroke. | IIb   | C     |

# Recommendations for atherothrombotic stroke prevention and treatment in patients with cerebrovascular disease or other ASCVD, and chronic kidney disease (3)

| Recommendations                                                                                                                                                                                                                            | Class | Level |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Anti-hypertensive therapy                                                                                                                                                                                                                  |       |       |
| A systolic blood pressure target of 120–129 mmHg is recommended in patients with CKD (who are not on KRT) and a history of stroke to reduce the risk of recurrent stroke.                                                                  | I     | C     |
| The maximum tolerated dose of an ACEI or ARB is recommended as the first-line blood pressure lowering agent in patients with CKD (including those on KRT) and a history of cerebrovascular disease for the secondary prevention of stroke. | I     | B1    |

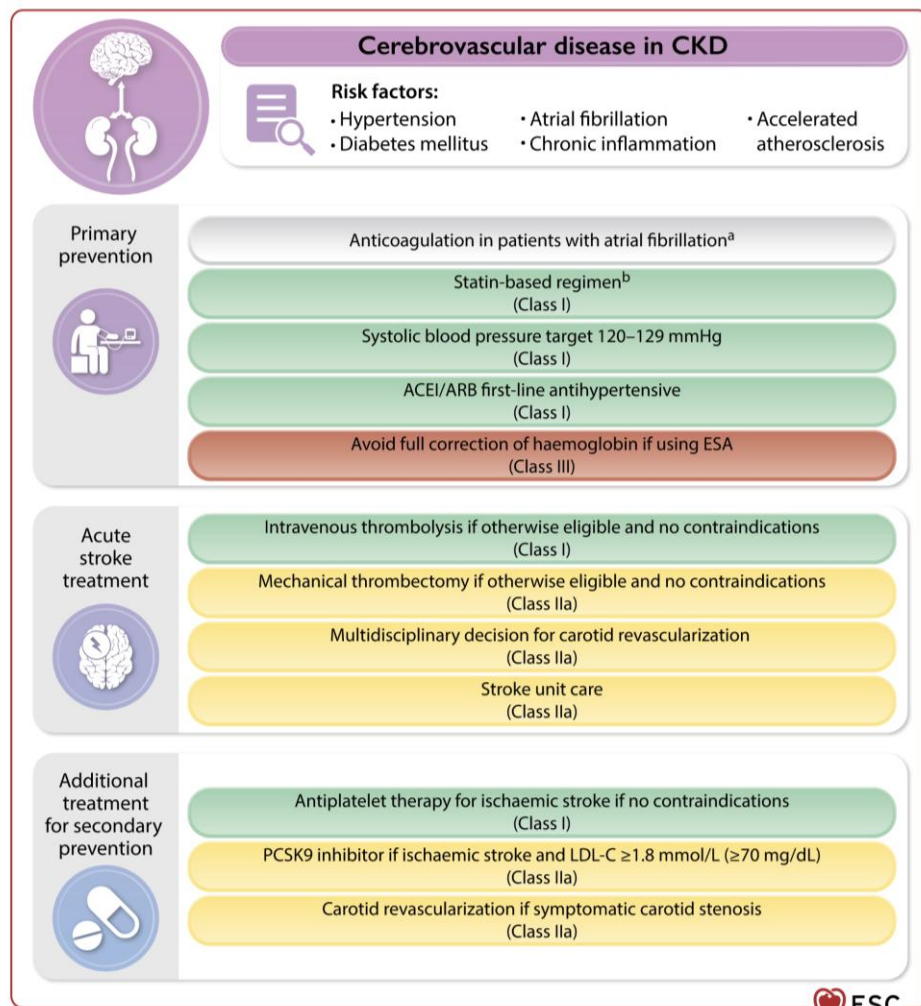
# Recommendations for atherothrombotic stroke prevention and treatment in patients with cerebrovascular disease or other ASCVD, and chronic kidney disease (4)

| Recommendations                                                                                                                                                                                                                         | Class      | Level     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Carotid revascularization procedures                                                                                                                                                                                                    |            |           |
| Carotid endarterectomy should be considered in patients with CKD who have symptomatic high-grade carotid stenosis to reduce the risk of stroke.                                                                                         | <b>IIa</b> | <b>B2</b> |
| Multidisciplinary team decision-making regarding carotid revascularization procedures should be considered in patients on KRT who have symptomatic carotid stenosis, given the higher peri-operative risk and lower long-term survival. | <b>IIa</b> | <b>C</b>  |
| Routine carotid revascularization is not recommended over optimal medical treatment in patients with CKD (including those on KRT) and asymptomatic carotid stenosis to reduce stroke risk.                                              | <b>III</b> | <b>B2</b> |
| Erythropoiesis-stimulating agents                                                                                                                                                                                                       |            |           |
| The use of erythropoiesis-stimulating agents to achieve full correction of anaemia is not recommended in patients with CKD and anaemia (including those on KRT) to avoid increased risk of stroke.                                      | <b>III</b> | <b>A</b>  |

# Recommendations for atherothrombotic stroke prevention and treatment in patients with cerebrovascular disease or other ASCVD, and chronic kidney disease (5)

| Recommendations                                                                                                                                                                                         | Class | Level |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Hyperacute stroke therapies and interventions                                                                                                                                                           |       |       |
| Treatment with intravenous thrombolysis is recommended in eligible patients with acute ischaemic stroke and CKD (including those on KRT) to reduce the risk of disability.                              | I     | B2    |
| Treatment with mechanical thrombectomy should be considered in eligible patients with large-vessel occlusive stroke and CKD (including those on KRT) to reduce the risk of disability.                  | Ila   | C     |
| Acute stroke unit admission after initial treatment should be considered in patients with acute ischemic stroke and CKD (including those on KRT) for further monitoring and rehabilitation post-stroke. | Ila   | C     |

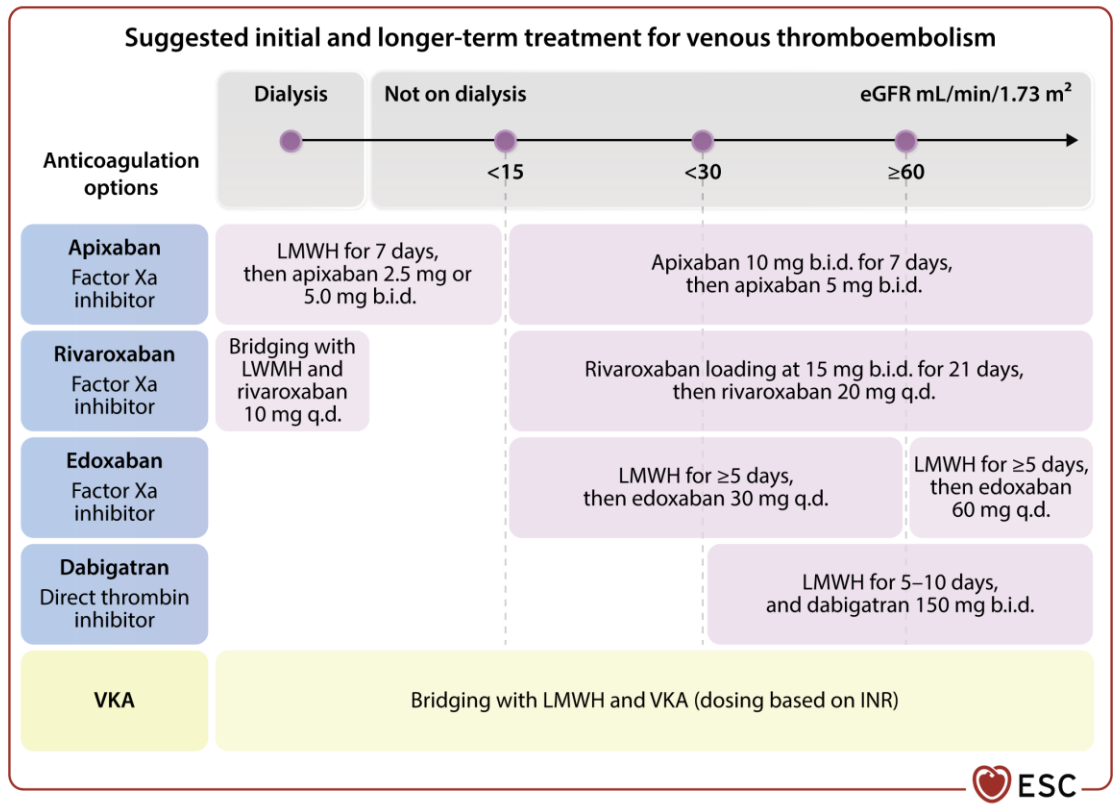
# Figure 15 Prevention and management of stroke in chronic kidney disease



# Recommendations for the treatment of venous thromboembolism in patients with chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                                                      | Class      | Level     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Anticoagulant treatment with DOACs (oral factor Xa or direct thrombin inhibitors) is recommended over treatment with LMWH and a VKA in patients with CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> for the treatment of VTE and prevention of recurrent VTE.                                 | <b>I</b>   | <b>B2</b> |
| Anticoagulant treatment with oral factor Xa inhibitors may be considered instead of conventional treatment with LMWH and a VKA in patients with CKD and an eGFR 15–29 mL/min/1.73 m <sup>2</sup> if the decision is made to anticoagulate for the treatment of VTE and prevention of recurrent VTE.  | <b>IIb</b> | <b>C</b>  |
| Anticoagulant treatment with either a VKA or low-dose apixaban (2.5 mg b.i.d) may be considered in patients with CKD and an eGFR $< 15$ mL/min/1.73 m <sup>2</sup> who are not receiving dialysis if the decision is made to anticoagulate for the treatment of VTE and prevention of recurrent VTE. | <b>IIb</b> | <b>C</b>  |
| Anticoagulant treatment with oral factor Xa inhibitors may be considered instead of conventional treatment with LMWH and a VKA in patients with CKD on dialysis if the decision is made to anticoagulate for the treatment of VTE and prevention of recurrent VTE.                                   | <b>IIb</b> | <b>C</b>  |

# Figure 16 Suggested initial and longer-term treatment for venous thromboembolism



# Recommendations for atrial fibrillation screening and risk prediction in chronic kidney disease

| Recommendations                                                                                                                                                                                                      | Class      | Level    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Opportunistic screening for AF by pulse taking or ECG may be considered in patients with CKD (including those on KRT) for earlier detection of AF.                                                                   | <b>IIb</b> | <b>C</b> |
| Existing ischaemic stroke risk scores are not recommended for use in patients with AF and CKD with an eGFR <30 mL/min/1.73 m <sup>2</sup> even if they include kidney parameters due to poor predictive performance. | <b>III</b> | <b>B</b> |
| Existing bleeding risk scores are not recommended for use in patients with AF and CKD with an eGFR <30 mL/min/1.73 m <sup>2</sup> even if they include kidney parameters due to poor predictive performance.         | <b>III</b> | <b>C</b> |

# Recommendations for anticoagulation for prevention of systemic embolism and stroke in patients with atrial fibrillation and chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                        | Class | Level |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Anticoagulation treatment with a DOAC is recommended in preference to a VKA in patients with AF and CKD with an eGFR $\geq 30$ mL/min/1.73 m <sup>2</sup> to reduce the risk of stroke, systemic thromboembolism, and bleeding.                        | I     | A     |
| Anticoagulation treatment with an oral factor Xa inhibitor should be considered in preference to a VKA in patients with AF and CKD with an eGFR 15–29 mL/min/1.73 m <sup>2</sup> to reduce the risk of stroke, systemic thromboembolism, and bleeding. | IIa   | B1    |
| Individualized clinical decision-making regarding anticoagulation should be considered for patients with AF and an eGFR $< 15$ mL/min/1.73 m <sup>2</sup> , including those on KRT, to balance the benefits and risks in this population.              | IIa   | C     |

# Recommendations for anticoagulation for prevention of systemic embolism and stroke in patients with atrial fibrillation and chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                                                                                                   | Class      | Level    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Anticoagulation with a VKA or apixaban at a reduced dose (2.5 mg b.i.d.) may be considered for patients with AF and an eGFR <15 mL/min/1.73 m <sup>2</sup> who are not receiving dialysis if the decision is made to anticoagulate to reduce the risk of stroke and systemic thromboembolism.                     | <b>IIb</b> | <b>C</b> |
| Anticoagulation with low-dose oral factor Xa inhibitors <sup>c</sup> may be considered preferentially over VKAs in patients with AF and an eGFR <15 mL/min/1.73 m <sup>2</sup> who are on dialysis if the decision is made to anticoagulate to reduce the risk of stroke, systemic thromboembolism, and bleeding. | <b>IIb</b> | <b>C</b> |

# Figure 17

## Recommendations on the use of anticoagulation therapy for patients with venous thromboembolism or atrial fibrillation in the context of chronic kidney disease

| Highly individualized clinical decision-making on decision to anticoagulate and choice of OAC                       |                                                                       |                                                     |                                                              |                                                                                      |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Treatment options when the decision is to anticoagulate                                                             |                                                                       |                                                     |                                                              |                                                                                      |
|                                                                                                                     | Dialysis                                                              | eGFR <15 mL/min/1.73 m <sup>2</sup> not on dialysis | eGFR 15–29 mL/min/1.73 m <sup>2</sup>                        | eGFR 30–59 mL/min/1.73 m <sup>2</sup>                                                |
| Patients with history of VTE<br>Treating VTE and preventing recurrent VTE                                           | Low-dose Factor Xa inhibitor preferred over LMWH or a VKA (Class IIb) | VKA or low-dose apixaban (Class IIb)                | Factor Xa inhibitor preferred over LMWH or a VKA (Class IIb) | DOAC (Factor Xa or direct thrombin inhibitor) preferred over LMWH or a VKA (Class I) |
| Patients with atrial fibrillation<br>Reducing risk of stroke, systemic thrombo-embolism, and bleeding complications | Low-dose Factor Xa inhibitor preferred over VKA (Class IIb)           | VKA or low-dose apixaban (Class IIb)                | Factor Xa inhibitor preferred over VKA (Class IIa)           | DOAC (Factor Xa or direct thrombin inhibitor) preferred over VKA (Class I)           |

# Recommendations for rate or rhythm control in patients with atrial fibrillation and chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                                                              | Class | Level |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| The choice of AAD is recommended to take into account the high prevalence of coronary and structural heart disease, and pro-arrhythmic effects (such as QT prolongation) that may be pronounced because of electrolyte imbalances in patients with AF and CKD to minimize pro-arrhythmic AAD-related events. | I     | C     |
| Rhythm control should be considered as an alternative to rate control in patients with AF and CKD to reduce symptoms.                                                                                                                                                                                        | IIa   | C     |
| Catheter ablation should be considered in patients on KRT with AF associated with haemodynamic instability during dialysis, to maintain sinus rhythm and improve the tolerability of dialysis.                                                                                                               | IIa   | C     |
| Catheter ablation may be considered in patients with AF, particularly paroxysmal AF, and CKD, as a rhythm-control strategy to reduce symptoms and avoid long-term adverse outcomes with AADs.                                                                                                                | IIb   | C     |

# Recommendations for device management considerations when treating patients with chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                                                                  | Class      | Level    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Application of an antibiotic envelope may be considered in addition to peri-operative antibiotic prophylaxis in patients on haemodialysis or after kidney transplantation who undergo CIED implantation to reduce the risk of CIED-related infection.                                                            | <b>IIb</b> | <b>C</b> |
| Where an ICD is indicated, implantation of a subcutaneous defibrillator may be considered as an alternative to transvenous ICD if there is no need for either bradycardia pacing, cardiac resynchronization, or ATP in patients with CKD on haemodialysis to reduce the risk of SCD and CIED-related infections. | <b>IIb</b> | <b>C</b> |
| Routine ICD implantation is not recommended in patients with CKD on haemodialysis with a LVEF $\geq 35\%$ for primary prevention of SCD.                                                                                                                                                                         | <b>III</b> | <b>C</b> |

# Recommendations for diagnosis and management of valvular heart disease in chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                                                | Class | Level |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Comprehensive TTE is recommended in patients with suspected valve disease and CKD irrespective of symptoms to evaluate the presence and severity of valvular heart disease and assess cardiovascular risk.                                                                                     | I     | C     |
| More frequent clinical and TTE re-evaluation (every 6–9 months) should be considered in patients with moderate to severe aortic stenosis and CKD with an eGFR <45 mL/min/1.73 m <sup>2</sup> to identify optimal timing for intervention due to unreliable symptoms and fast progression rate. | IIa   | C     |
| Clinical and TTE evaluation should be considered after initiating dialysis in patients with known moderate to severe valvular heart disease to identify optimal timing of intervention due to haemodynamic changes during dialysis.                                                            | IIa   | C     |
| Transcatheter aortic valve intervention instead of surgical replacement should be considered in patients with severe aortic stenosis with CKD aged ≥70 years due to the less invasive procedure, lower risk of complications, and similar outcomes over 3-5 years.                             | IIa   | C     |

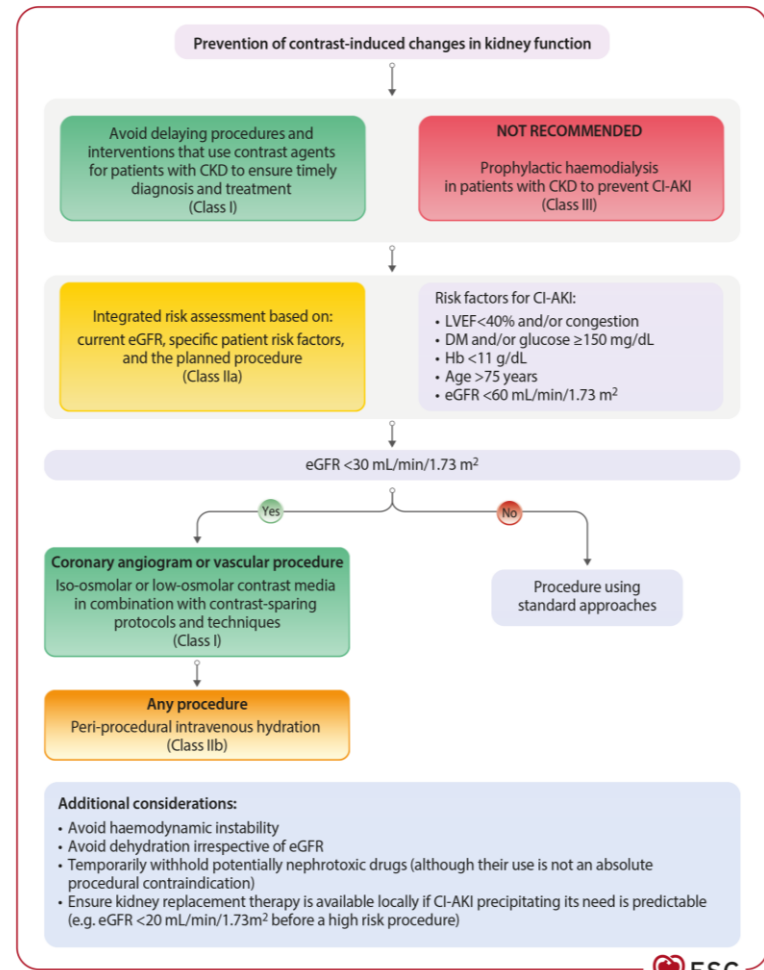
# Recommendations to support safe use of contrast in chronic kidney disease (1)

| Recommendations                                                                                                                                                                                                                                                                                      | Class | Level |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Avoiding delays in procedures and interventions that use contrast agents is recommended for patients with CKD to ensure timely diagnosis and treatment.                                                                                                                                              | I     | C     |
| Use of iso-osmolar or low-osmolar iodinated contrast media in combination with contrast-sparing protocols and techniques is recommended in patients with an eGFR <30 mL/min/1.73 m <sup>2</sup> not on dialysis undergoing coronary angiography or vascular procedures to reduce the risk of CI-AKI. | I     | A     |
| Use of group two or three gadolinium-based contrast agents is recommended in patients with an eGFR <30 mL/min/1.73 m <sup>2</sup> for MRI scans requiring such contrast due to the low risk of toxicity (nephrogenic systemic fibrosis).                                                             | I     | B2    |

## Recommendations to support safe use of contrast in chronic kidney disease (2)

| Recommendations                                                                                                                                                                                                                        | Class      | Level     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| An integrated assessment of risk based on current eGFR, specific patient risk factors, as well as the planned procedure should be considered to identify patients with CKD at high risk of harm associated with CI-AKI.                | <b>IIa</b> | <b>C</b>  |
| Peri-procedural intravenous hydration may be considered in patients with an eGFR <30 mL/min/1.73 m <sup>2</sup> not on dialysis, undergoing any procedure requiring administration of iodinated contrast to reduce the risk of CI-AKI. | <b>IIb</b> | <b>C</b>  |
| Prophylactic haemodialysis is not recommended in patients with CKD to prevent CI-AKI, as this has been associated with harm.                                                                                                           | <b>III</b> | <b>B2</b> |

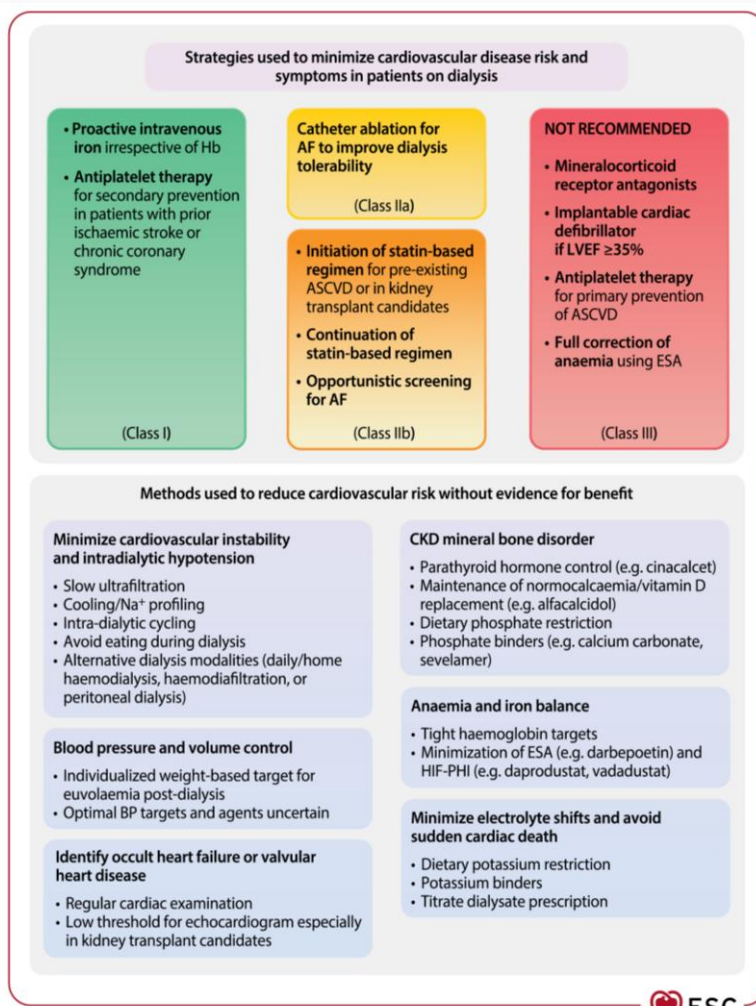
# Figure 18 Algorithm to support safe use of contrast in chronic kidney disease



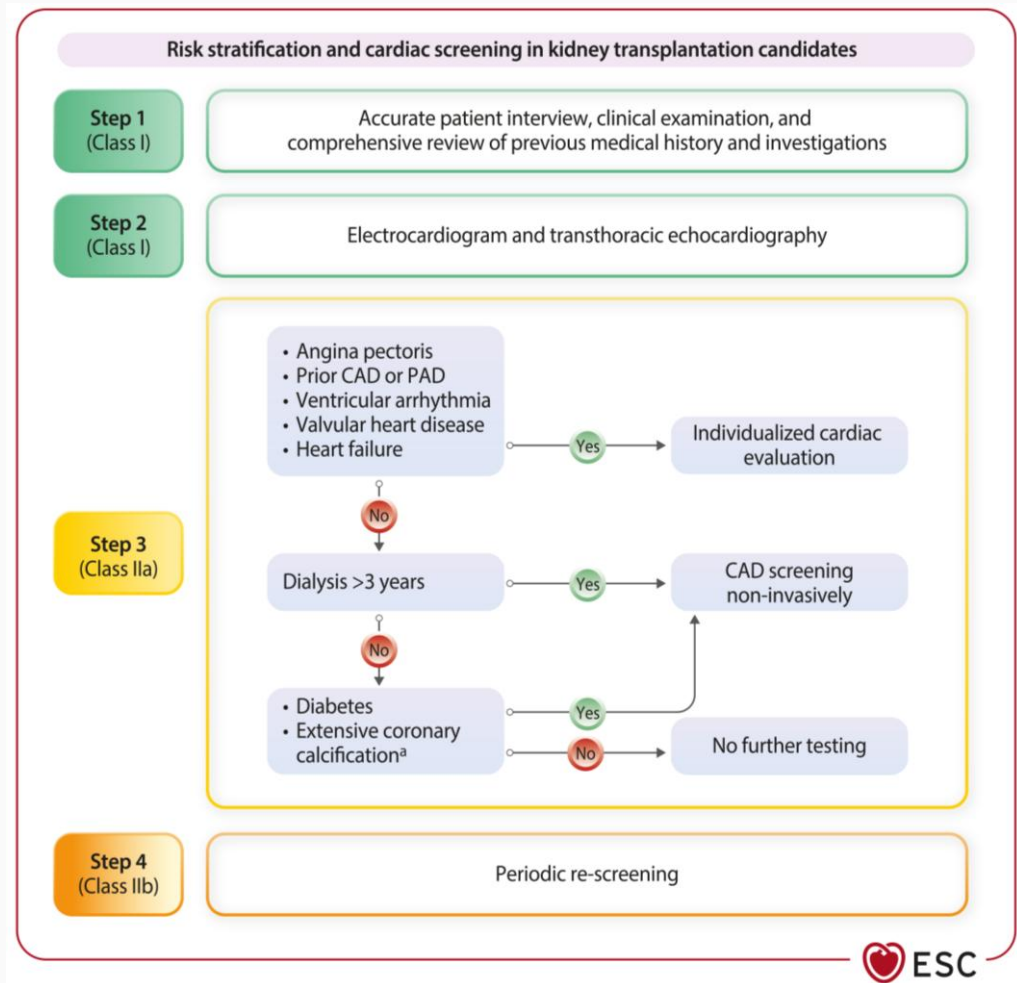
# Recommendations for prevention of cardiovascular disease in patients on dialysis

| Recommendations                                                                                                                                                                                                                                                                     | Class      | Level     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Proactive regular high-dose intravenous iron (e.g. iron sucrose) is recommended in patients on maintenance haemodialysis, unless the serum ferritin concentration is >700 µg/L or transferrin saturation is ≥40%, to reduce risk of cardiovascular events, including heart failure. | <b>I</b>   | <b>B1</b> |
| Routine use of MRAs is not recommended in patients on dialysis because of the risk of serious hyperkalaemia without evidence of clear benefit on cardiovascular death or hospitalizations for heart failure.                                                                        | <b>III</b> | <b>A</b>  |

# Figure 19 Strategies used to minimise risk of cardiovascular disease and symptoms in patients on dialysis



# Figure 20 Risk stratification and cardiac screening in kidney transplant candidates



# Recommendations for cardiac evaluation of kidney transplant candidates (1)

| Recommendations                                                                                                                                                                                                                                    | Class | Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Patient interview, clinical examination, and comprehensive review of previous medical history are recommended in all kidney transplant candidates with the aim of stratifying cardiac risk.                                                        | I     | C     |
| Resting electrocardiogram and transthoracic echocardiography are recommended in kidney transplant candidates with the aim of identifying valvular disease, and to assess ventricular structure and function.                                       | I     | C     |
| Individualized decision-making (including multidisciplinary discussion) regarding revascularization is recommended in asymptomatic kidney transplant candidates with established obstructive CAD to minimize risk associated with transplantation. | I     | B2    |

## Recommendations for cardiac evaluation of kidney transplant candidates (2)

| Recommendations                                                                                                                                                                                                                                                                                                   | Class      | Level    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Individualized cardiac evaluation and potential multidisciplinary team discussion should be considered in kidney transplant candidates with angina, established atherosclerotic disease, ventricular arrhythmia, valvular heart disease or HF to identify the optimal pre-, peri-, and post-operative management. | <b>IIa</b> | <b>C</b> |
| Myocardial perfusion imaging test or anatomic testing with coronary CT angiography should be considered in high-risk asymptomatic kidney transplant candidates with the aim of risk stratifying, diagnosing obstructive CAD, and guiding treatment of modifiable risk factors and revascularization.              | <b>IIa</b> | <b>B</b> |
| Using coronary artery calcification findings from previous chest CT scans, if available, should be considered in kidney transplant candidates with the aim of enhancing risk stratification and guiding treatment of modifiable risk factors for CAD.                                                             | <b>IIa</b> | <b>C</b> |

# Recommendations for cardiac evaluation of kidney transplant candidates

| Recommendations                                                                                                                                                                                                                                                                    | Class      | Level    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Periodic re-screening of CAD may be considered in kidney transplant candidates on the waiting list with either symptoms, earlier abnormalities or new unexpected (cardiac) findings to reduce risks of complications from revascularization and delayed access to transplantation. | <b>IIb</b> | <b>C</b> |
| Pre-operative CAD screening using diagnostic testing is not recommended in kidney transplant candidates with low probability of obstructive CAD due to risk of potential harms of screening including delaying listing suitable candidates.                                        | <b>III</b> | <b>C</b> |

# Recommendations for management of cardiovascular disease risk in kidney transplant recipients

| Recommendations                                                                                                                                                | Class | Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| A systolic blood pressure target of 120–129 mmHg is recommended in kidney transplant recipients to improve cardiovascular outcomes.                            | I     | C     |
| Calcium channel blockers should be considered in kidney transplant recipients to lower blood pressure to an individual's target and reduce risk of graft loss. | IIa   | B2    |
| SGLT2 inhibitors may be considered in kidney transplant recipients with either pre- or post-transplant diabetes to reduce cardiovascular risk.                 | IIb   | C     |
| GLP-1RAs may be considered in kidney transplant recipients with either pre- or post-transplant diabetes to reduce cardiovascular risk.                         | IIb   | C     |

# Recommendation for multidisciplinary care in chronic kidney disease

| Recommendations                                                                                                                                                                                                                                                                    | Class      | Level    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| Involvement of relevant nephrology team members in CVD multidisciplinary care teams should be considered to support patients with CKD with an eGFR <30 mL/min/1.73 m <sup>2</sup> or on KRT to ensure CKD-related considerations are included in shared decision-making processes. | <b>Ila</b> | <b>C</b> |

# Figure 21 Person-centred care in cardiovascular disease and chronic kidney disease

